

Università degli Studi di Padova

Fintrack
Personal Finance Management App

Team WASP

Rajmonda Bardhi Meviola Karriqi Giacomo Masiero Danil Smirnov

Academic Year 2025 / 2026

Presentation Outline

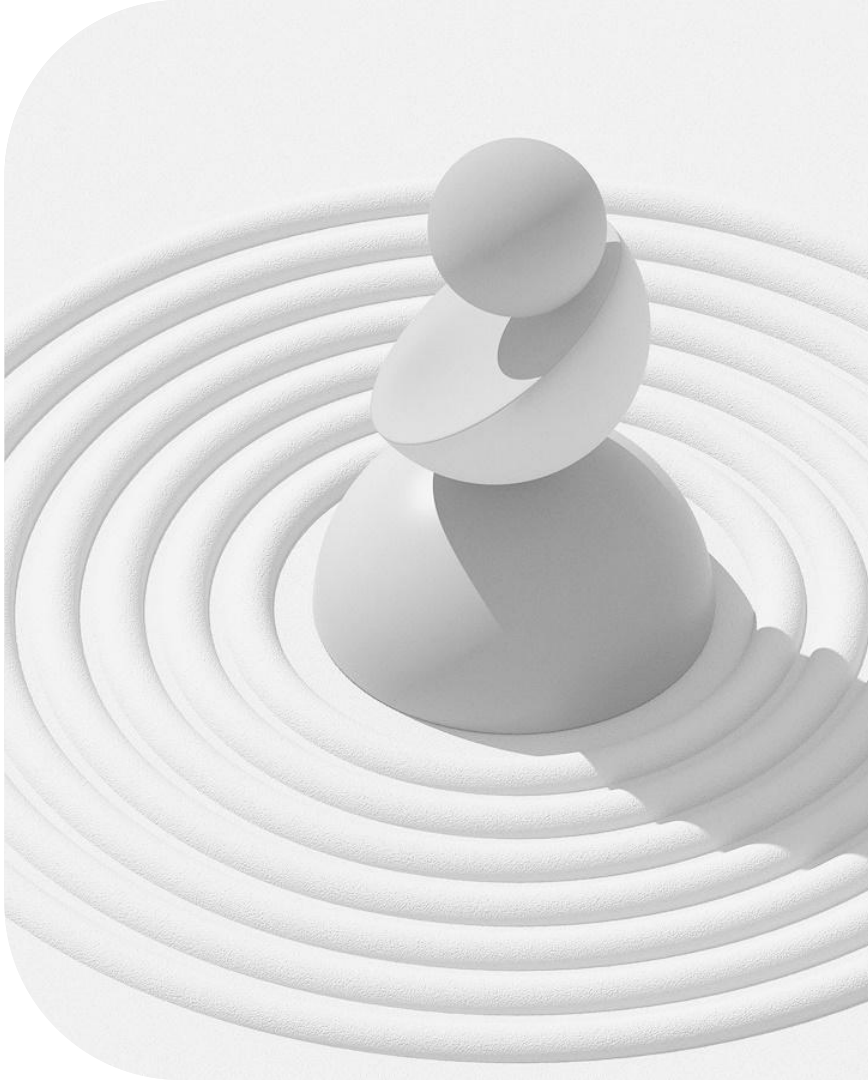
1. The problem and the objective
2. The BackEnd
3. The FrontEnd

The Problem and the Vision

The Problem: Users often manage their wealth fragmented across multiple accounts, cards, and wallets

The Solution: Provide a single, unified dashboard

The Objective: Create a platform that aggregates all "wallets" into a single, comprehensive overview



Personal Finance: Objective

Monitoring and Visibility

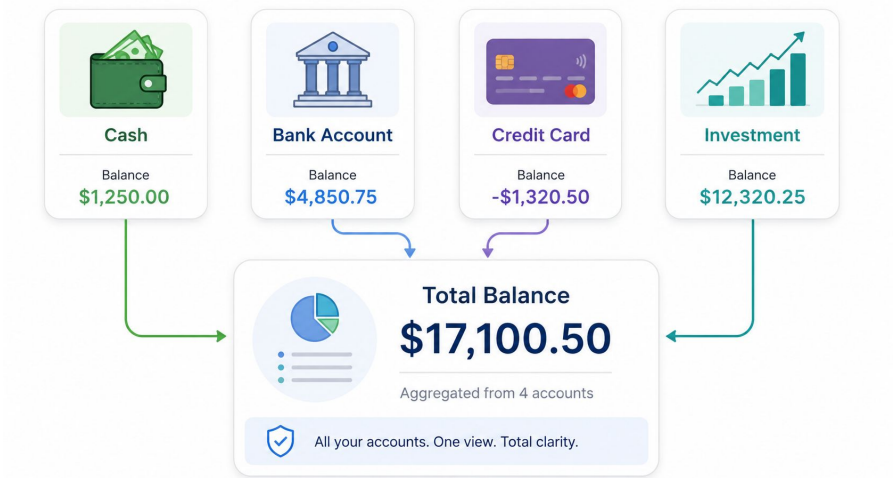
Long Term Planning

Forecasting

Objective: Monitoring and Visibility

Aggregation of balances from multiple accounts

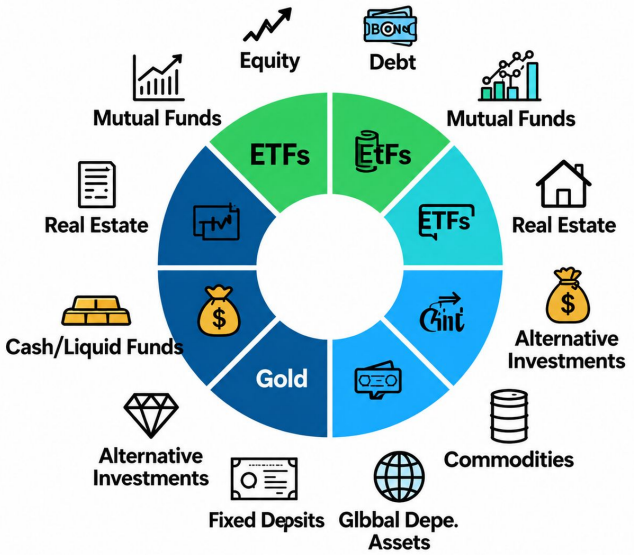
Immediate visualization of Net Balance, Year-to-Date (YTD) Total Expenses, and Savings



Objective: Long-Term Planning

Introduction of PAC (Piano di Accumulo Capitale) plans.

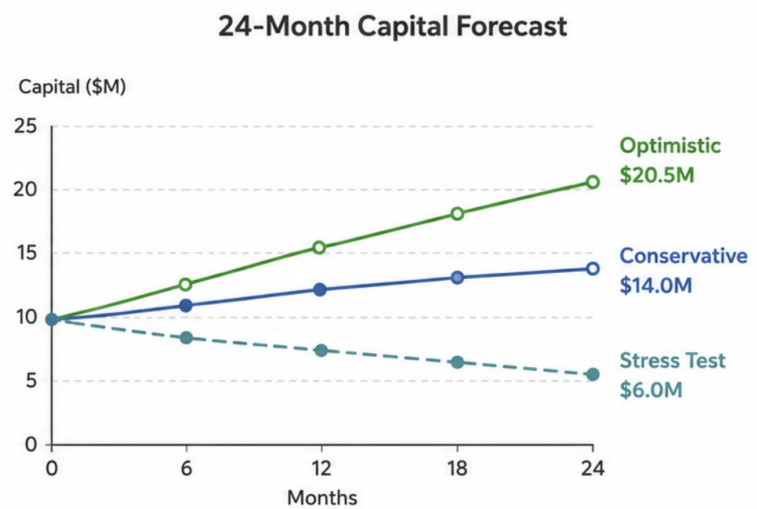
The system allows the user to schedule repeated monthly investments towards specific assets and track their impact on the budget.



Objective: Forecasting

Analytical module to forecast capital for the next 24 months.

It works based on three different macroeconomic scenarios



Tools













Three-Tier MVC Architecture

Presentation Logic Layer:

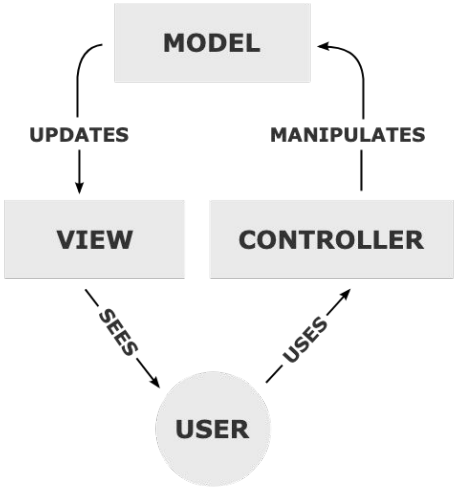
Web interface served to the client

Business Logic Layer:

The core of the system inside Tomcat (Servlets, Filters, Services)

Data Logic Layer:

Persistence management on the DB (DAOs, PostgreSQL)

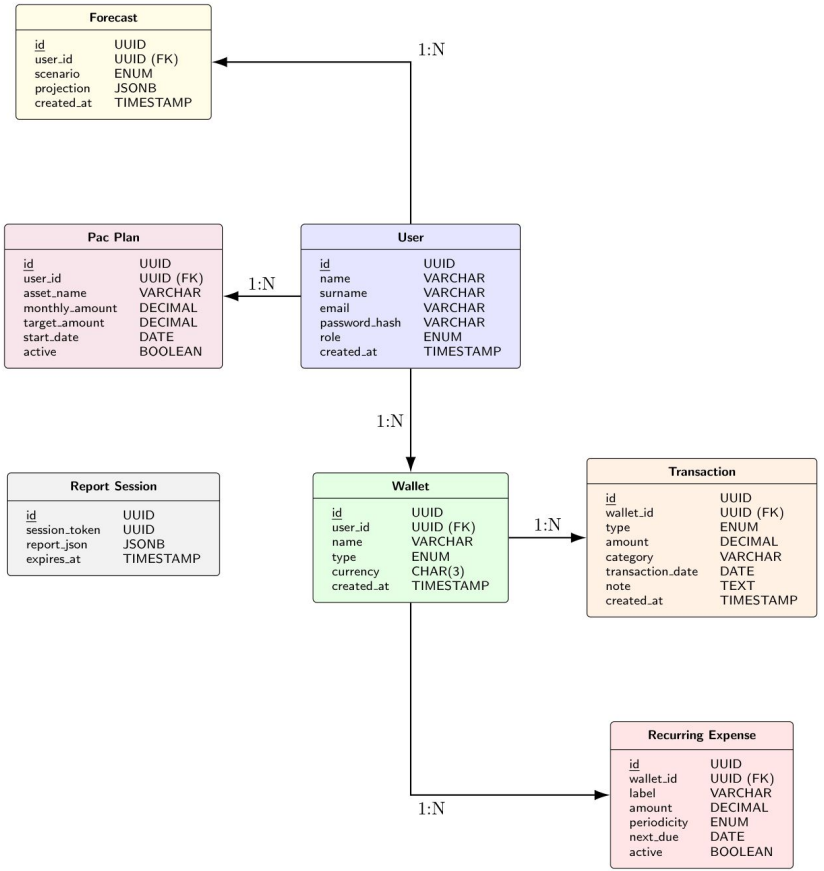


The Backend

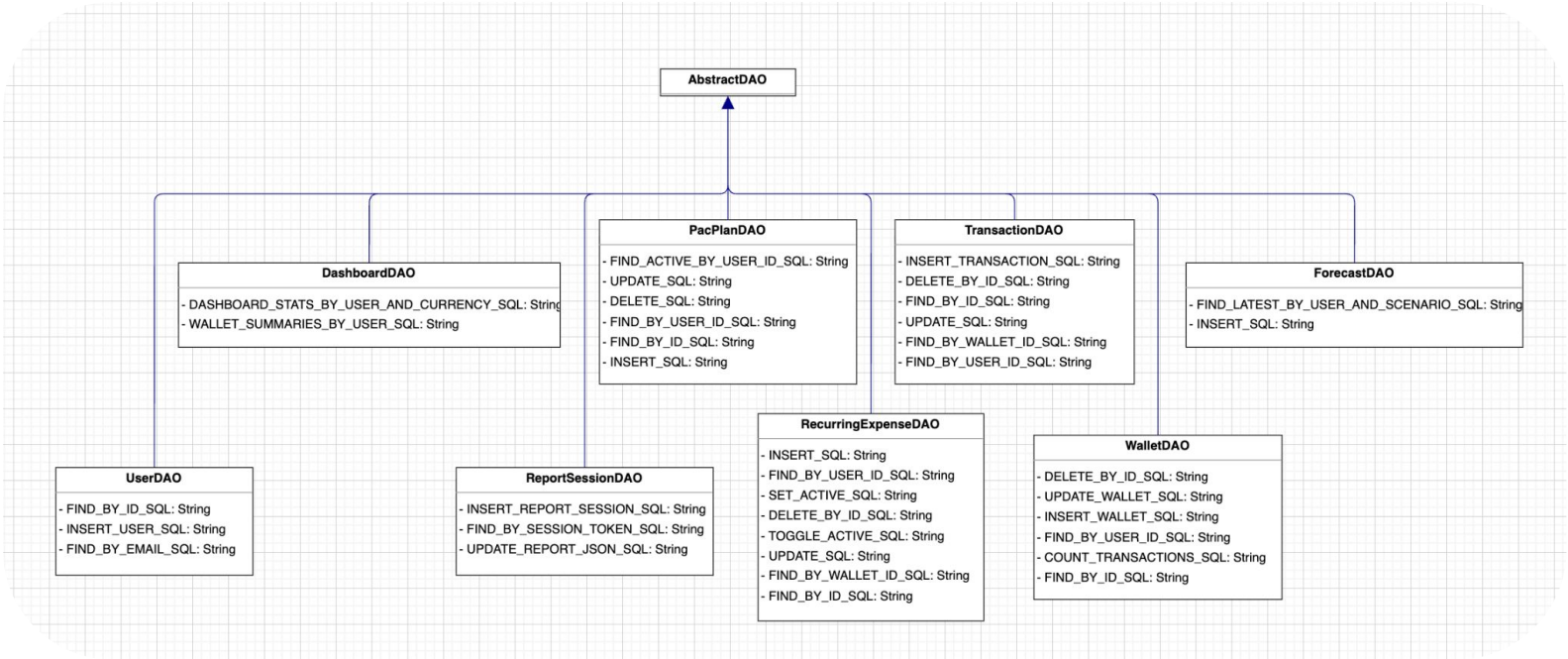
Database Schema

UUIDv4 Primary Keys: No sequential IDs

JSONB Support: Flexible storage for Forecasting and Guest sessions

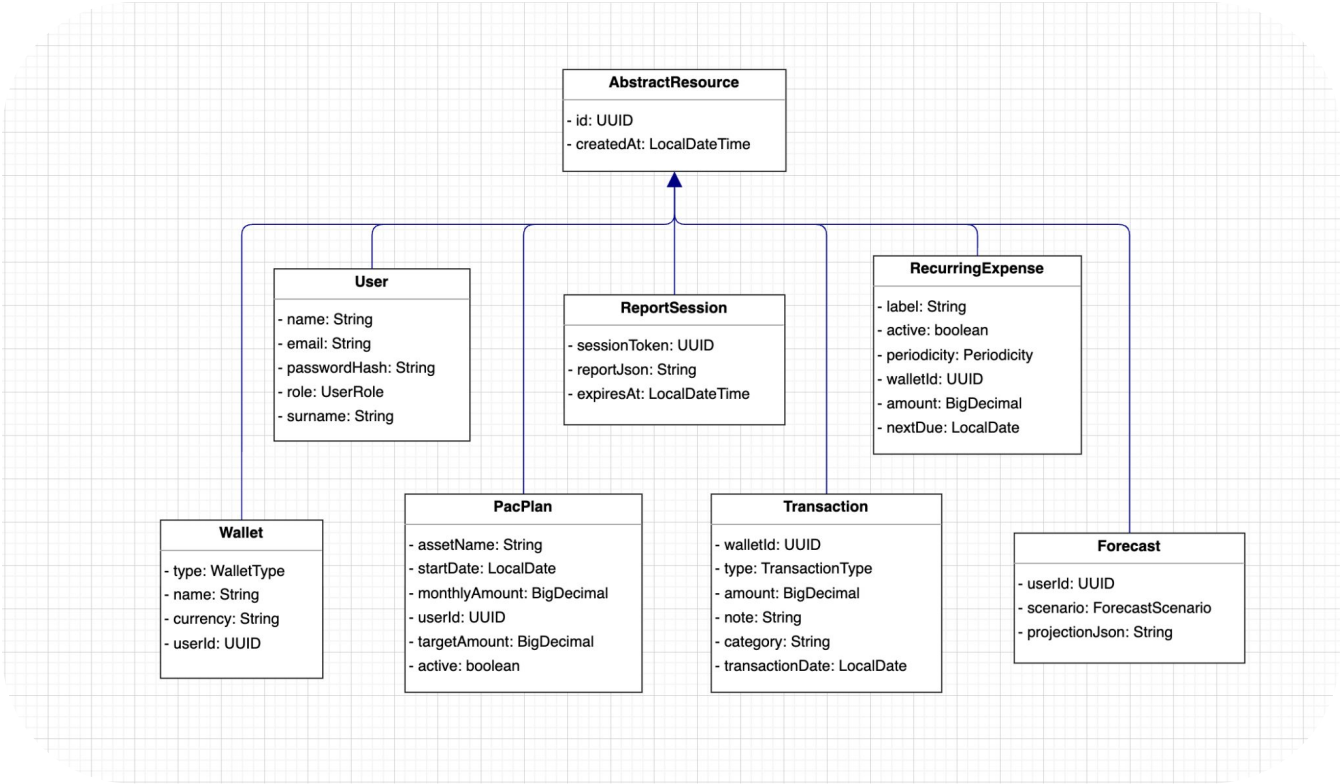


DAOs



SQL Injection Protection: only PreparedStatements with bound parameters (?) are used

Resources

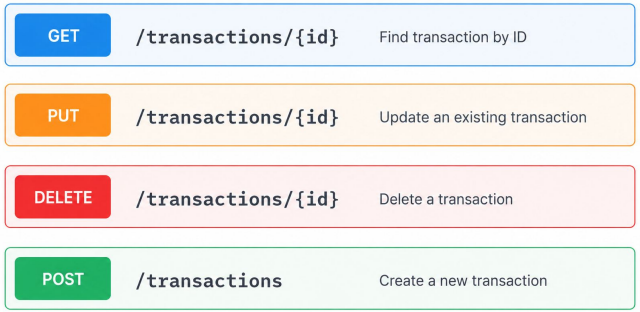

AbstractResource
- id: UUID
- createdAt: LocalDateTime
User
- name: String
- email: String
- passwordHash: String
- role: UserRole
- surname: String
ReportSession
- sessionToken: UUID
- reportJson: String
- expiresAt: LocalDateTime
RecurringExpense
- label: String
- active: boolean
- periodicity: Periodicity
- walletId: UUID
- amount: BigDecimal
- nextDue: LocalDate
Wallet
- type: WalletType
- name: String
- currency: String
- userId: UUID
PacPlan
- assetName: String
- startDate: LocalDate
- monthlyAmount: BigDecimal
- userId: UUID
- targetAmount: BigDecimal
- active: boolean
Transaction
- walletId: UUID
- type: TransactionType
- amount: BigDecimal
- note: String
- category: String
- transactionDate: LocalDate
Forecast
- userId: UUID
- scenario: ForecastScenario
- projectionJson: String

RESTful API Calls

The application exposes true API endpoints under the /api/* path

No HTML: These endpoints exchange exclusively JSON payloads

They utilize semantic HTTP methods



Security: Password Protection

Plaintext passwords are never sent to the database

A key derivation algorithm resistant to brute-force attacks is used: PBKDF2

During login, the hash recalculated in memory is compared with the secured one in the database

```
    public static String hashPassword(final String plainPassword) {
        if (plainPassword == null || plainPassword.isBlank()) {
            throw new IllegalArgumentException("Password cannot be null or
blank.")}

        byte[] salt = new byte[SALT_LENGTH];
        new SecureRandom().nextBytes(salt);

        byte[] hash = pbkdf2(plainPassword.toCharArray(), salt);

        String saltBase64 = Base64.getEncoder().encodeToString(salt);
        String hashBase64 = Base64.getEncoder().encodeToString(hash);

        return ITERATIONS + ":" + saltBase64 + ":" + hashBase64;
    }
```

Security: Authentication Filter

It intercepts every request and checks if the HttpSession has the authenticatedUser attribute set

If the user is not authenticated

302 Redirect to /login

if navigating via a browser

401 Unauthorized

if it was an AJAX call

```
@Override
    protected void doFilter(final HttpServletRequest request,
                            final HttpServletResponse response,
                            final FilterChain chain) throws IOException, ServletException {

        HttpSession session = request.getSession(false);
        boolean authenticated = session != null && session.getAttribute("authenticatedUser") != null;

        if (!authenticated) {
            String path = request.getServletPath();
            logger.log(Level.FINE, "Unauthenticated access attempt to {0} from {1}",
                    new Object[]{path, request.getRemoteAddr()});
            if (path != null && path.startsWith("/api/")) {
                response.setContentType("application/json");
                response.setCharacterEncoding("UTF-8");
                response.setStatus(HttpServletResponse.SC_UNAUTHORIZED);
                response.getWriter().write("{\"error\":\"Authentication required.\"}");
            } else {
                response.sendRedirect(request.getContextPath() + "/login");
            }
            return;
        }

        chain.doFilter(request, response);
    }
```

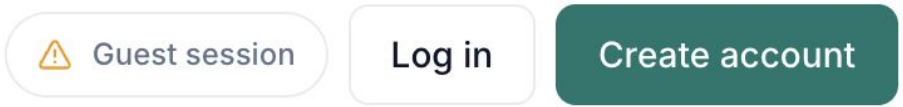
Guest Section: Overview

Goal: Allow visitors to test core financial tools with zero friction.

Available Calculators: Loan Amortization, Compound Interest, and Budget Analysis.

No Account Required: Completely accessible from the Home page without providing any personal data.

Guest Session Filter: If a visitor enters the calculators section, the filter checks if they possess an identifier



Report Export



Not just on-screen reading, the project natively exposes the interchange of tangible documents

The module hooks the external Open Source library Apache PDFBox

Composes the tabular report with immediate saving via the user's browser

The FrontEnd

The Presentation Layer: Main Areas

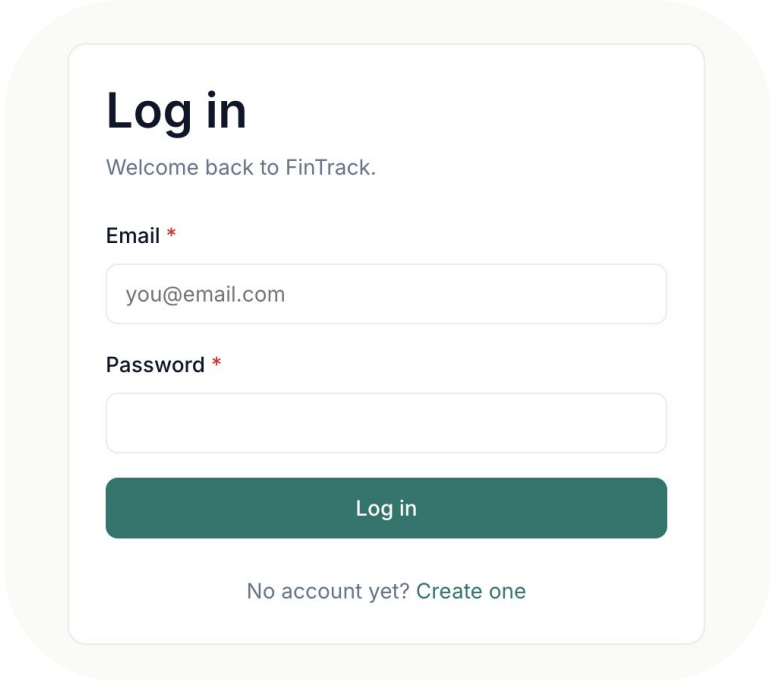
The application divides the user interface into three isolated macro-areas:

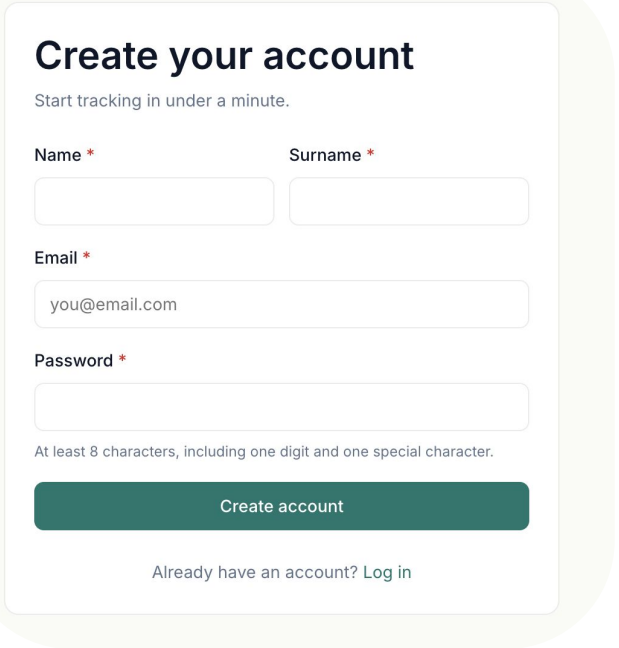
Public Flow: Home, Login, and Registration pages

Authenticated Flow: Protected main dashboard, Wallet Management, and Projections

Guest Flow: Public calculators area and temporary reports

Login and Registration



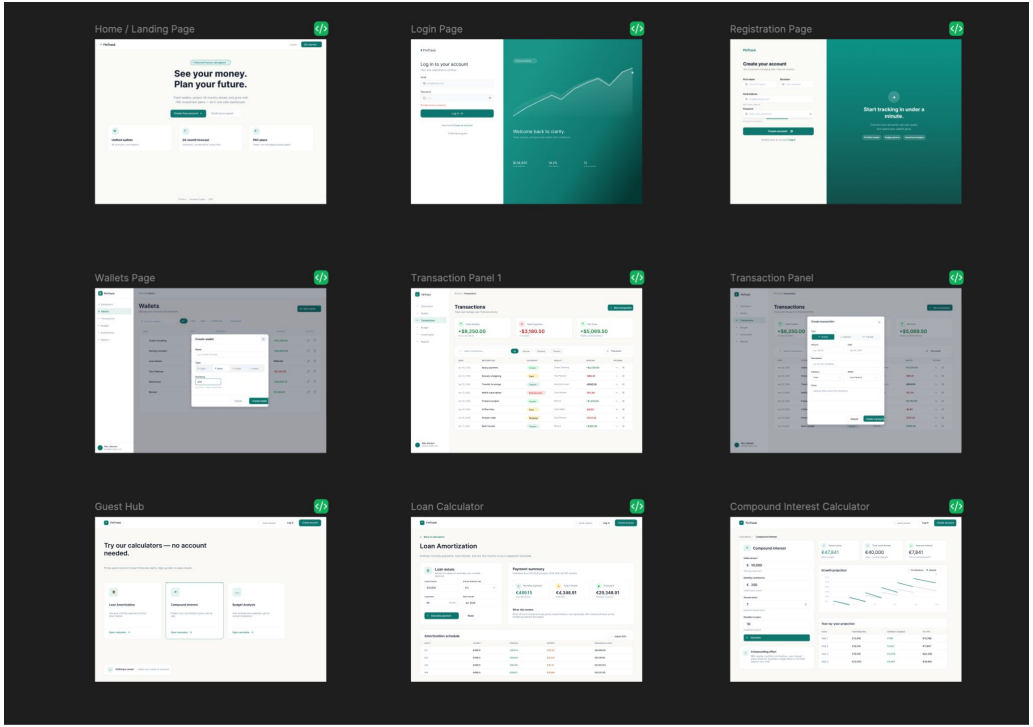


Design

Figma

Font family - Inter





Our Palette

#0F172A

#0F766E

#64748B

#E0F0EF

#FAFAF7

#DC2626

#E5E7EB

#16A34A

#D97706

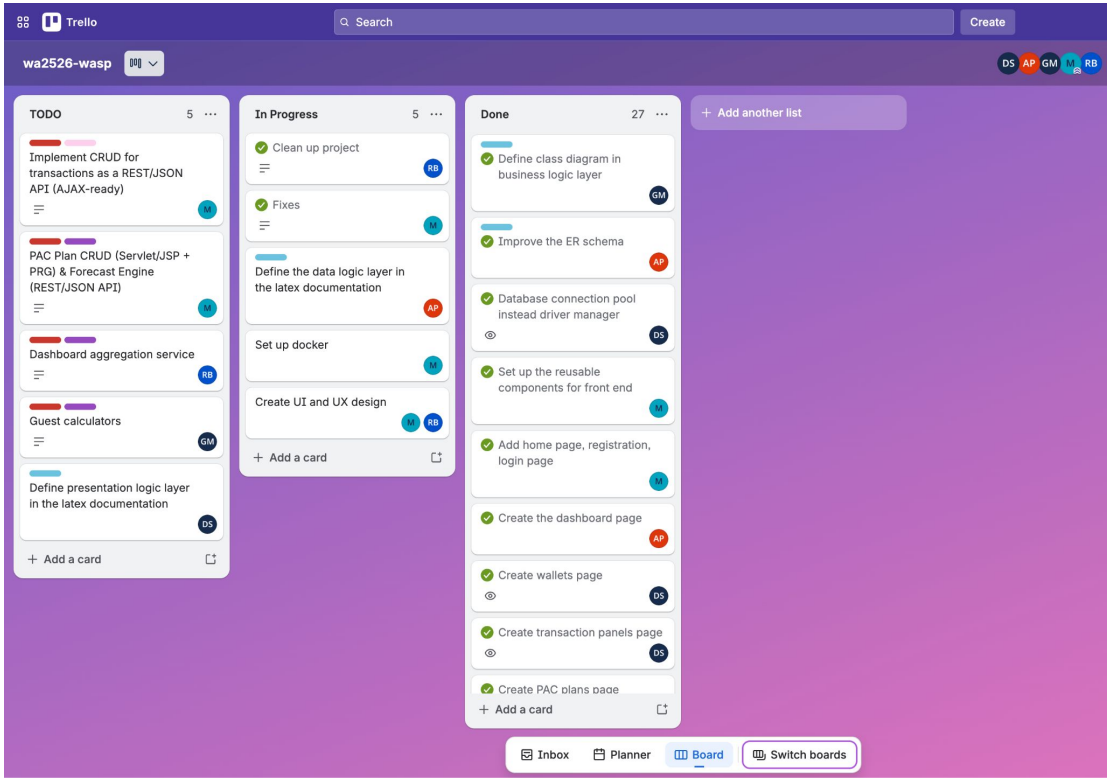
#7C3AED

#FEF3C7

#B45309

Team Workflow

Trello



Live Demo

Thank You